COMPUTER SCIENCE

O Level Notes

Syllabus 2210 / 0478

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O Level Computer Science

Chapter 3: Hardware

3.1 Computer Architecture · 3.2 Input and Output Devices · 3.3 Data Storage · 3.4 Network Hardware

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1. The Central Processing Unit (CPU)

The **CPU** (also called the **processor** or **microprocessor**) carries out **all the instructions and data processing** in a computer. It is usually a single **integrated circuit** (microchip).

The CPU contains:
- **Control Unit (CU)**
- **Arithmetic and Logic Unit (ALU)**
- **Registers** (connected to the rest of the computer by **buses**)

1.1. Von Neumann Architecture

Introduced in the mid-1940s as the **stored program computer**. Its main features:
- The computer has a **CPU** (processor).
- The CPU can **access memory directly**.
- Memory stores **both programs and data**.
- Stored programs are made of **instructions** that are executed **one after another** (sequentially).

1.2. Components of the CPU

Component	Function
ALU	Carries out **arithmetic** (+, −, shifting) and **logic** (AND, OR, NOT) operations. Multiplication and division are done by repeated addition, subtraction and shifts.
Control Unit	**Reads instructions** from memory, **decodes** them, and sends **control signals** along the control bus to the other components. Keeps data flow **synchronised**.
System clock	Produces **timing signals** on the control bus so all operations are synchronised. Without it the computer would crash.
Registers	Very small, very fast memory locations **inside the CPU** (see below).

RAM (Immediate Access Store): holds the programs and data the CPU is using. It is much **faster** than backing store (HDD/SSD), so the CPU copies what it needs from backing store into RAM.

1.3. Special Purpose Registers

Register	Short	Function
Program Counter	PC	Stores the **address of the next instruction** to be fetched
Memory Address Register	MAR	Stores the **address** of the memory location currently being read from or written to
Memory Data Register	MDR	Stores the **data/instruction** just read from memory, or about to be written to memory

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Current Instruction Register	CIR	Stores the **instruction** currently being decoded and executed
Accumulator	ACC	Temporarily stores the **results of ALU calculations**

Reading from memory: (1) the address is placed in the **MAR**; (2) a **read signal** is sent; (3) the contents of that location are placed in the **MDR**.

Writing to memory: (1) the data is placed in the **MDR**; (2) the address is placed in the **MAR**; (3) a **write signal** is sent and the data is stored at that address.

> **Exam Tip:** PC = **where next**. MAR = **which address**. MDR = **what data**. CIR = **current instruction**. ACC = **calculation result**.

1.4. System Buses

A **bus** is a set of parallel wires; **each wire carries one bit**.

Bus	Direction	What it carries
Address bus	Unidirectional (one way)	**Addresses** from the CPU to memory (one way so addresses never come back to the CPU)
Data bus	Bidirectional (both ways)	**Data, instructions and addresses** between CPU, memory and input/output devices
Control bus	Bidirectional (both ways)	**Control signals** from the CU to all other components (usually 8 bits wide)

- **Wider address bus** ⟶ more memory locations can be addressed. 16 bits ⟶ 2^{16} = 65 536 locations. 32 bits ⟶ 4 294 967 296 locations.
- **Wider data bus** ⟶ larger **word length** (bits moved at once) ⟶ better performance.

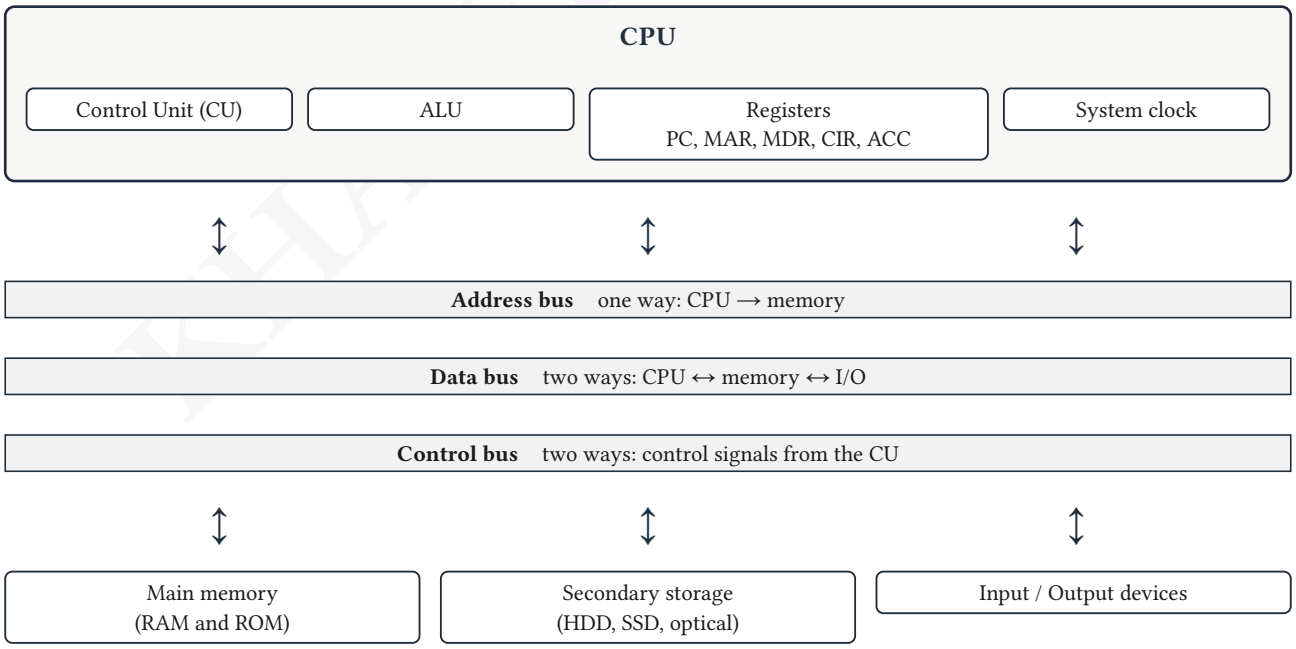


Figure 1: Von Neumann architecture: the CPU, memory, input/output devices and the three system buses

2. Fetch–Decode–Execute Cycle

The CPU repeats this cycle for every instruction.

Fetch

1. The **PC** holds the address of the next instruction.
2. This address is copied from the PC to the **MAR** (using the **address bus**).
3. The instruction at that address is copied into the **MDR** (using the **data bus**).
4. The contents of the MDR are copied to the **CIR**.
5. The **PC is incremented by 1** so it points to the next instruction.

Decode

1. The **control unit decodes** the instruction in the CIR.

Execute

1. The decoded instruction is carried out by sending **control signals** (via the control bus) to the correct components. Calculations are done in the **ALU** and stored in the **ACC**.

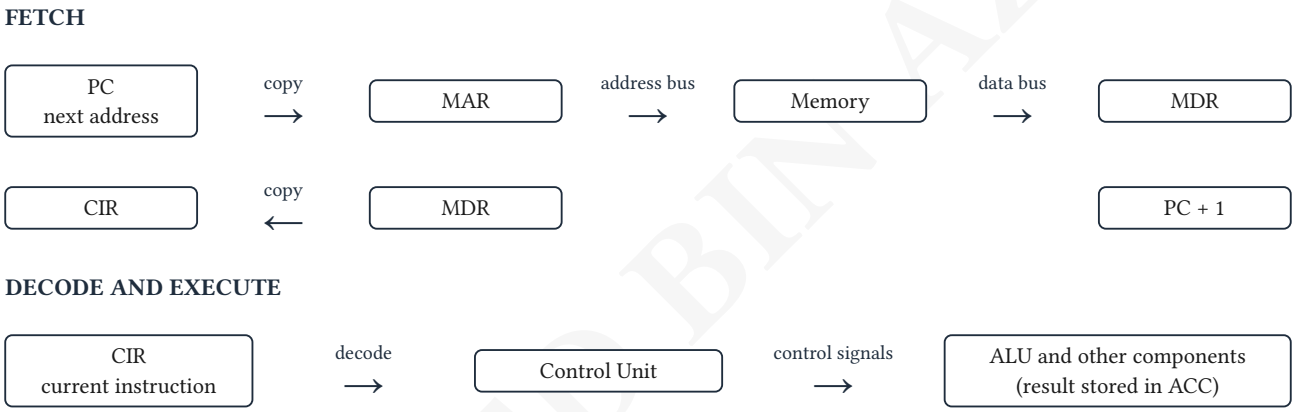


Figure 2: The fetch–decode–execute cycle: how an instruction moves between the registers

> **Exam Tip:** Order to remember: PC ⟶ MAR ⟶ MDR ⟶ CIR, then PC + 1. Do not forget that the **PC is incremented during the fetch stage**.

3. CPU Performance

Factor	Effect on performance
Clock speed	Number of clock cycles per second (e.g. 3.5 GHz = 3.5 billion cycles per second). **Higher clock speed ⟶ more instructions per second.**
Number of cores	A **core** = ALU + CU + registers. More cores (dual = 2, quad = 4) ⟶ more instructions processed at the same time. **Doubling the cores does not double the speed** because the cores must communicate with each other.
Cache	Very fast memory (SRAM) **inside the CPU** holding **frequently used instructions and data**. The CPU checks the cache first, then RAM. **Larger cache ⟶ faster CPU.**
Bus width	Wider address and data buses ⟶ more data and more memory locations handled at once.

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Overclocking: running the CPU at a clock speed **higher than the manufacturer's setting** (changed in the BIOS). Risks:

- Instructions cannot finish in time → **unsynchronised operation**, crashes and instability.
- The CPU **overheats** → unreliable performance and damage.

4. Instruction Set

An **instruction** is made of an **opcode** and an **operand**.

- **Opcode:** tells the CPU **what operation** to do (e.g. ADD, JMP, LDA).
- **Operand:** the **data** (or the address/register holding the data) to be acted on.



Figure 3: Structure of a machine code instruction

The **instruction set** is the **complete list of machine code instructions** that a particular CPU can understand (for example x86).

> Instruction set ≠ program code. High-level program code must be translated by a **compiler** or **interpreter** into instructions the CPU understands.

5. Embedded Systems

An **embedded system** is a **combination of hardware and software designed to do one specific set of tasks**, built into a larger device.

A **general-purpose computer is not an embedded system** because it can do many different tasks.

Type	Description
Microprocessor	An integrated circuit with **only a CPU**. RAM, ROM and other parts must be added.
Microcontroller	A **CPU + RAM + ROM + I/O ports** on a single chip.
System on Chip (SoC)	A complete system (CPU, memory, I/O, storage) on **one chip**; may include a microcontroller.

How it works: input (from a **sensor** or the user, e.g. a keypad) → **processing** by the microcontroller → **output** (e.g. actuator, display, alarm). Embedded systems are **feedback** systems and work in **real time**.

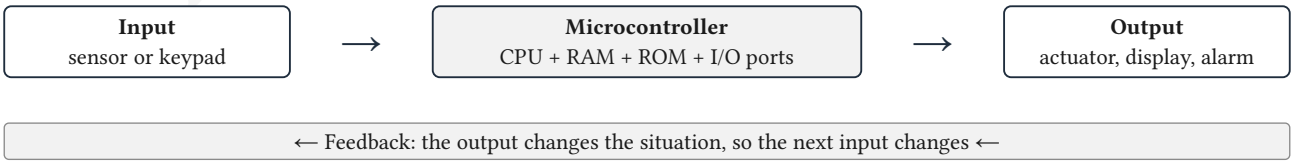


Figure 4: Block diagram of an embedded system

Examples: car engine/ABS/airbags/GPS, set-top box, security alarm, office lighting, vending machine, washing machine, microwave, digital camera.

Benefits	Drawbacks
Small and easy to fit into devices	Hard to **upgrade** to new technology

Low cost to make	**Troubleshooting** needs a specialist
Dedicated to one task, so a **simple interface** and often no operating system	If connected to the internet, open to **hackers and viruses**
Use **very little power**	Often **thrown away** instead of repaired (wasteful)
Fast reaction to changing input	A "simple" interface (one knob) can actually be **confusing**
Can be **controlled remotely** (e.g. by phone)	

6. Input Devices

6.1. Barcode Scanner

A **barcode** is a series of **dark and light parallel lines of varying thickness** representing digits.

How it is scanned:

1. A **red laser or red LED** is shone across the barcode.
2. **Dark bars reflect little or no light**; light bars reflect light.
3. The reflected light is read by **sensors (photoelectric cells)**.
4. The pattern of reflections is converted into **digital data**.



red laser / LED sweeps across the bars



Figure 5: A barcode being scanned and the data flow at the checkout

What happens at a supermarket checkout:

1. The barcode number is looked up in the **stock database** (it is the **key field**).
2. The **price and details** are sent back to the checkout (POS terminal).
3. The **stock level is reduced by 1**.
4. If the stock level is **less than or equal to the re-order level**, more stock is **automatically ordered** (a flag stops it re-ordering again).

Benefits: faster checkout, fewer pricing errors, itemised bill, automatic stock control, easy price changes, up-to-date sales information.

6.2. QR Codes

A **Quick Response (QR) code** is a **matrix of dark and light squares** (pixels) that can be read by a **smartphone camera**.

- Holds far more data than a barcode (up to **4296 characters** or 7089 digits; a barcode holds about 30 digits).
- The **three large squares** in the corners help the reader find the **position and angle** of the code.
- **Reading:** the phone camera takes an image ⟶ a QR app converts the squares into data ⟶ the browser follows any web address, or opens a phone number, or stores a boarding pass.

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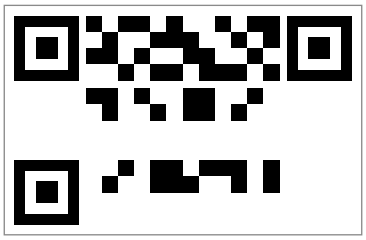


Figure 6: Layout of a QR code: three corner squares show the position and angle

Advantages over barcodes	Disadvantages
Holds **much more data**	**More than one QR format** exists
Has **built-in error correction** (works even if damaged)	Can contain **malicious links or code** that takes the user to a fake website or installs a virus
Read by a **phone camera**, no special scanner	
Easy to send as text or image; can be **encrypted**	

6.3. Digital Camera

1. Light passes through the **lens** onto a **CCD** (a grid of millions of light sensors, each one a **pixel**).
2. Each sensor produces an **electric charge** depending on the light.
3. An **ADC (analogue to digital converter)** changes each charge into a **binary value** (brightness and colour).
4. Most cameras use **24-bit RGB**: 8 bits each for red, green and blue (0–255 each).



Figure 7: How a digital camera captures an image

More pixels → better image quality and larger file size.

6.4. Keyboard

1. Pressing a key **completes a circuit** on the circuit board under the keys.
2. The CPU identifies **which key** was pressed.
3. The CPU looks up the character in an index and uses its **ASCII value**.

Keyboards are **slow and error-prone** for data entry, and long use can cause **repetitive strain injury (RSI)**. **Ergonomic keyboards** reduce this.

6.5. Microphone

1. Sound waves make the **diaphragm vibrate**.
2. The diaphragm moves a **coil of wire** past a **permanent magnet**, producing an **analogue electric current**.
3. The **sound card (ADC)** converts it into **digital data**.



Figure 8: From sound waves to digital data in a microphone

6.6. Optical Mouse

1. A **red LED** shines light onto the surface.
2. A **CMOS sensor** takes about **1500 images per second** of the reflected light.
3. A **digital signal processor (DSP)** compares the images to work out the **coordinates** of the movement.
4. The computer moves the **cursor** to those coordinates.

Advantages over a mechanical mouse: no moving parts (more reliable), dirt cannot get trapped, works on almost any surface.

6.7. 2D and 3D Scanners

2D scanner (inputs **paper documents** and photos):

1. The document is placed on glass and lit by a **bright lamp**.
2. A **scan head** moves across the page and the reflected light goes through **mirrors and a lens** onto a **CCD**.
3. The CCD converts the light into **electric charge** and software builds a **digital image**.

OCR (optical character recognition) software converts the scanned image of text into **editable text**. Used, for example, to read **passports** at airports.

3D scanner: scans a **solid object** and records its **x, y and z coordinates** to create a **3D digital model**. The model can be used in **CAD** or sent to a **3D printer**. A **CT scanner** builds a 3D image from many thin 2D **slices**.

6.8. Touch Screens

Type	How the touch position is found	Advantages	Disadvantages
Capacitive	A finger **changes the electrostatic field** of the conductive layer. A microcontroller calculates the coordinates.	Clear image, durable, **multi-touch** (projective)	Needs a **bare finger** or special stylus; affected by electromagnetic interference
Infrared	A grid of **infrared beams** crosses the screen. A touch **breaks the beams**, and the sensors give the position.	Multi-touch, durable, works with a cracked screen	Sensitive to **moisture and light**; can be triggered accidentally
Resistive	Two **resistive layers** touch when pressed. The **change in voltage** gives the position.	Resists dust and water; works with **gloves or stylus**	Low sensitivity, **no multi-touch**, poor in sunlight, scratches easily

7. Output Devices

7.1. Actuators

An **actuator** is an **output device that converts an electrical signal into movement**. Examples: **motor, relay, solenoid** (e.g. opening a valve, moving a conveyor belt). A **DAC** is needed if the actuator requires an analogue signal.

7.2. Light Projectors

DLP (Digital Light Projector)	LCD Projector
Uses a **DMD chip** with millions of **tiny mirrors**; each mirror tilts ON or OFF to make a light or dark pixel	White light is split into **red, green, blue** by mirrors, passed through **three LCD panels** and recombined by a prism
Higher contrast, smaller, lighter, quieter, longer life	Sharper image, better colour saturation, more energy efficient
Colour is not as good; can show shadows on moving images	Lower contrast; limited life (panels degrade and turn yellow)

7.3. Inkjet Printer

Parts: print head with nozzles, ink cartridges, stepper motor and belt, paper feed.

Droplet technology:
- **Thermal bubble:** a tiny **resistor heats the ink** into a bubble that **pushes a droplet** onto the paper.
- **Piezoelectric:** a **crystal vibrates** when charged and **forces ink out**.

Printing process:
1. Data is sent to the **printer driver**, which converts it to a format the printer understands.
2. The driver **checks the printer is available** (on, not busy, ink, paper).
3. Data is stored in the **printer buffer**.
4. Paper is fed in; the **print head moves side to side** spraying ink; the paper advances after each pass.
5. When the buffer is empty, the printer sends an **interrupt** to the CPU asking for more data.

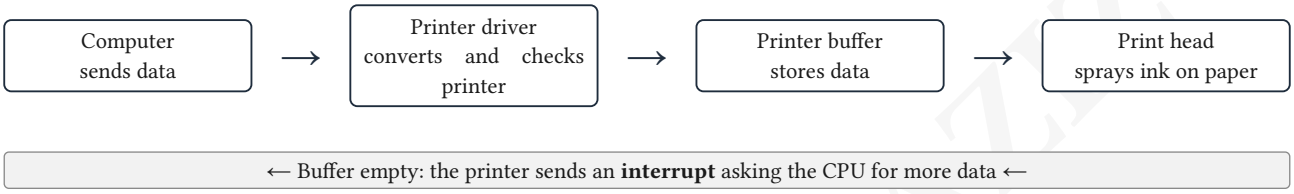


Figure 9: Data flow when printing with an inkjet printer

7.4. Laser Printer

Uses **dry powder (toner)** and **static electricity**. The **whole page** is formed at once.

1. The **drum is given a charge**.
2. A **laser scans the drum**, removing the charge where the text/image will be, so the page image is formed in charge.
3. **Toner sticks** to the correct areas of the drum.
4. **Paper with an opposite charge** is rolled over the drum and the toner transfers to it.
5. The paper's charge is removed so it does not stick to the drum.
6. A **fuser** (heated rollers) **melts the toner** onto the paper.
7. A **discharge lamp** clears the drum for the next page.



Figure 10: The stages of laser printing

Inkjet	Laser
Few pages, **good-quality colour**, photos	**High volume**, fast, many copies (flyers, posters, office documents)
Small ink cartridges and paper tray	Large toner cartridges and paper trays
Cheaper printer, ink is costly per page	Dearer printer, cheaper per page

7.5. 3D Printer

Builds a **solid object layer by layer** (**additive manufacturing**), using materials such as resin, powdered metal, ceramic or paper. Normal machining **removes** material (**subtractive**).

Process: design in **CAD** → software prepares the file → printer builds **layer by layer** (about 0.1 mm each) → object is cleaned and cured.

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Figure 11: Stages of 3D printing

Types: direct 3D printing (inkjet-type head moves in x, y, z) and **binder 3D printing** (a powder layer, then a binder/glue sprayed on it).

Uses: prosthetic limbs, surgical models, aerospace parts, spare parts for old machines, prototypes, fashion and art.

7.6. Screens: LED, LCD and OLED

Screen	Key points
LED screen	Made of tiny **red, green and blue LEDs** that **produce their own light**. Used for large outdoor displays.
LCD screen	**Liquid crystals** form the pixels but **do not produce light**, so a **backlight** is needed. Modern screens use **LED backlighting** (these are called "LED TVs" but are really LCD screens). Older screens used **CCFL** tubes.
OLED screen	**Organic** layers **give off their own light**, so **no backlight** is needed.

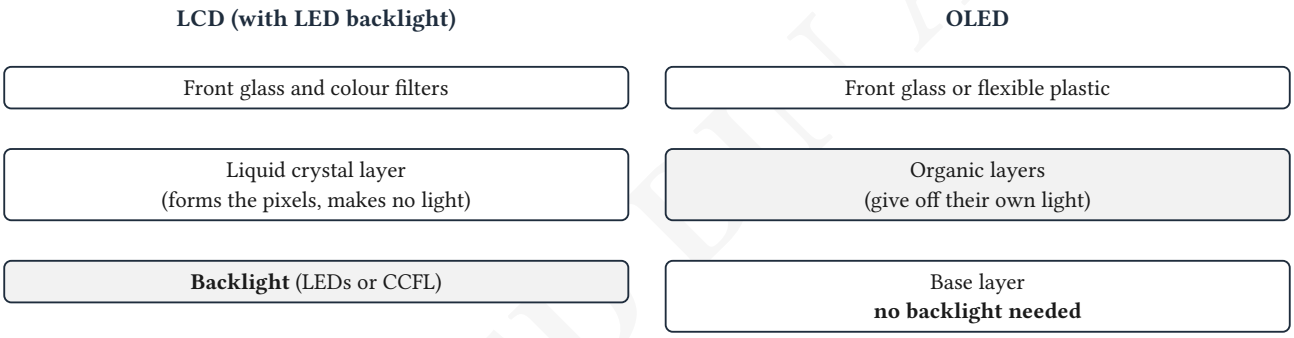


Figure 12: Layers of an LCD screen compared with an OLED screen

Advantages of LED backlight over CCFL: reaches full brightness immediately, whiter and more vivid, thinner screens, longer life, **less power and heat**.

Advantages of OLED over LED/LCD: thinner, lighter and flexible (can be curved or folded), **no backlight**, so **less power**, brighter, **wider viewing angle** (about 170°).

7.7. Loudspeaker

1. Digital sound data goes through a **DAC**, which converts it into an **electric current**.
2. An **amplifier** makes the current strong enough.
3. The current flows through a **coil** next to a **permanent magnet**, making the coil and **cone vibrate**, which produces **sound waves**.



Figure 13: How a loudspeaker produces sound

8. Sensors

A **sensor** is an **input device that measures a physical property** of its surroundings. Real data is **analogue**, so the signal goes through an **ADC (analogue to digital converter)** before the computer can process it. Sensors send readings **continuously**; the **microprocessor** analyses them.

Sensor	Measures	Example use
Temperature	Heat/temperature	Central heating, greenhouse, chemical process
Moisture	Water level in soil or a material	Greenhouse soil, food factory
Humidity	Water vapour in the air	Greenhouse, microchip factory, buildings
Light	Brightness of light	Street lights, car headlights switching on automatically
Infrared (active)	A broken infrared beam	Intruder alarm, automatic wipers
Infrared (passive)	Heat given off by objects	Intruder alarm (body heat), freezer temperature
Pressure	Pressure or weight	Weighing lorries, burglar alarm pad, reactor gas pressure
Acoustic (sound)	Sound	Burglar alarm (footsteps, breaking glass), detecting leaks
Gas	Oxygen / carbon dioxide level	Air pollution, greenhouse, car exhaust
pH	Acidity	Greenhouse soil, chemical process
Magnetic field	Change in magnetic field	Anti-lock brakes (ABS) in cars
Accelerometer	Acceleration or sudden change in speed	Airbags, phone screen rotation
Proximity	Presence of a nearby object	Phone screen switches off at your ear
Flow	Flow rate of a liquid or gas	Gas pipes, hospital respiratory devices
Level	Height of a liquid or powder	Car fuel tank, tablet production

8.1. Monitoring and Control

Monitoring system	Control system
Microprocessor **only watches** the readings	Microprocessor **changes the process** by controlling devices
If a reading is outside the range: **warning or alarm** (screen, siren)	If a reading is outside the range: sends signals to **actuators** (heater, valve, motor, pump)
Does not affect what is being measured	Output **affects the next input**: this is **feedback**
Examples: hospital patient monitoring, burglar alarm, river pollution	Examples: central heating, street lights, greenhouse, ABS, chemical process

General control process:

1. Sensors send **analogue** readings.
2. The **ADC** converts them to **digital**.
3. The microprocessor **compares** the data with the **pre-set (stored) value**.
4. If action is needed, it sends a signal (through a **DAC** if needed) to an **actuator**.
5. The **sensors read again**, and the loop repeats (**feedback**).

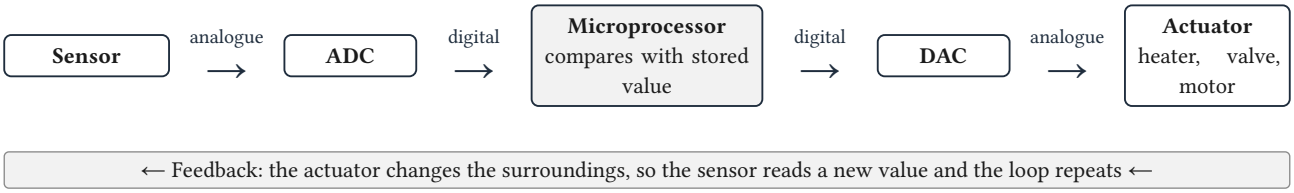


Figure 14: A computer control system with feedback

Example: central heating. The temperature sensor sends readings → ADC → microprocessor compares with the pre-set temperature. If the temperature is **lower** than the pre-set value, the microprocessor switches on the **gas valve and water pump** (via a DAC). If it is **equal or higher**, no action is taken. This repeats until the system is switched off.

Example: street light. A light sensor sends readings → ADC → microprocessor. If the reading is **below the stored value**, the lamp is switched **on**; otherwise it is switched **off**. The reading is checked again after a delay (e.g. 30 minutes) to stop flickering.

> **Exam Tip:** In control answers always mention: **sensor → ADC → microprocessor compares with stored value → signal (DAC) → actuator → sensor reads again**.

9. Primary Memory and Secondary Storage

Primary memory	Secondary storage
Directly accessed by the CPU	**Not** directly accessed by the CPU
RAM, ROM (and cache)	HDD, SSD, CD/DVD/Blu-ray, USB flash drive
Fast, smaller capacity	**Slower**, much larger capacity
RAM is **volatile**; ROM is non-volatile	**Non-volatile** (keeps data without power)

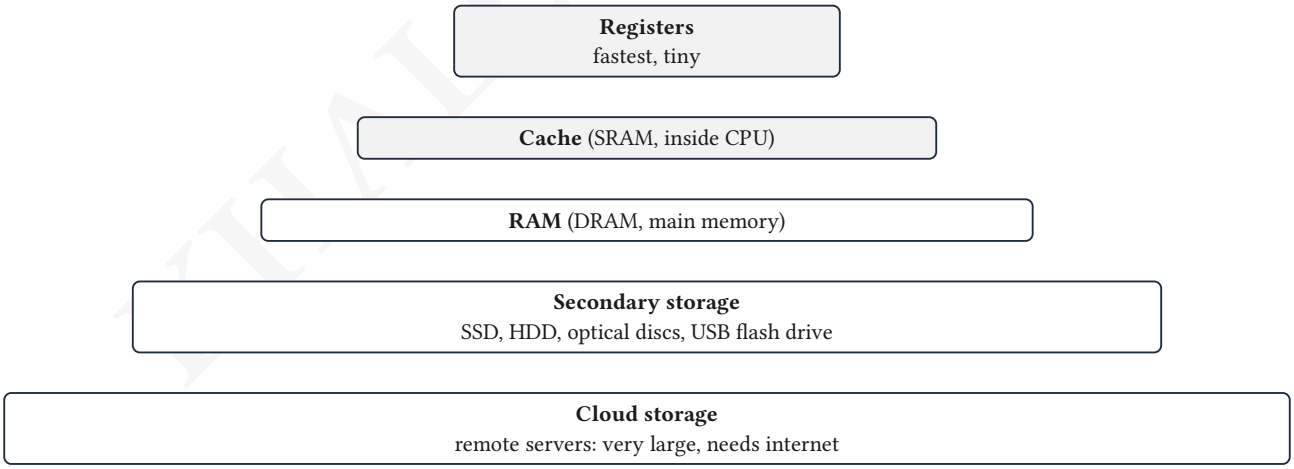


Figure 15: Memory hierarchy: faster, smaller and dearer towards the top; slower, larger and cheaper towards the bottom

9.1. RAM and ROM

RAM (Random Access Memory)	ROM (Read-Only Memory)
Volatile: contents lost when power is off	**Non-volatile**: contents kept without power
Read and write	**Read only** (cannot be changed)

Stores **programs, data and part of the OS currently in use**	Stores the **start-up instructions (BIOS / bootstrap)**
Temporary	Permanent
More RAM ⟶ computer generally runs faster	Size is fixed

Example, remote-controlled toy car: ROM stores the start-up routine, factory settings and fixed routines (what each button does). **RAM** stores the instructions received from the controller and any user-programmed routines.

9.2. DRAM and SRAM

DRAM (Dynamic RAM)	SRAM (Static RAM)
Transistors and capacitors	Uses **flip-flops**
Must be **constantly refreshed**	**Does not need refreshing**
Cheaper, higher capacity, less power	**Faster** access, more expensive, more power
Used for **main memory**	Used for **CPU cache**

10. Secondary Storage

10.1. Magnetic: Hard Disk Drive (HDD)

- Data is stored as **magnetic patterns** on spinning **platters**, in **tracks and sectors**.
- **Read-write heads** (electromagnets) move across the platters to read and write data.
- **Latency:** the time for the required sector to **rotate round to the read-write head**. HDDs are slow because of latency and head movement.
- **Fragmentation:** over time a file's sectors become spread out, which **slows access**. **Defragmentation software** tidies the sectors.
- **Removable HDDs** connect by USB and are used for **backup** and moving files.

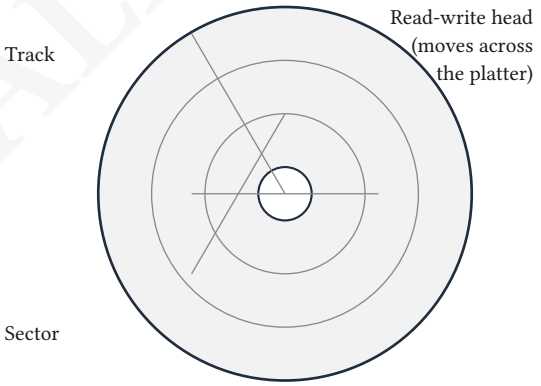


Figure 16: Top view of a hard disk platter: tracks, sectors and the read-write head

10.2. Solid State: SSD and USB Flash Drive

- **No moving parts.** Data is stored by **trapping electrons in floating-gate transistors** (NAND or NOR flash). A charged floating gate = **1**, uncharged = **0**.
- **Non-volatile**, rewritable memory. Data must be **erased before it is rewritten**.

Advantages over HDD: more **reliable** (no moving parts), **faster** access, **lighter**, thinner, **lower power**, runs **cooler**, no start-up delay, silent. These make SSDs suitable for **laptops**.

Drawbacks: limited write cycles (SSD endurance) and **higher cost per GB**.

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USB flash drive: small, portable, used to **transfer and back up files**. It can also act as a **dongle** (a key needed to run protected software).

10.3. Optical: CD, DVD and Blu-ray

- Data is stored as **pits and lands** on a **single spiral track**, from the centre outwards.
- A **laser** reads (and writes) the data. **R** = write once; **RW** = rewritable.
- The **shorter the laser wavelength**, the **smaller the pits** and the **greater the capacity**.

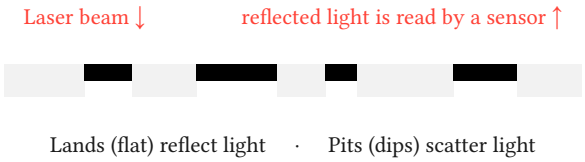


Figure 17: How a laser reads pits and lands on an optical disc

	CD	DVD	Blu-ray
Laser colour	Red	Red	**Blue-violet**
Wavelength	780 nm	650 nm	**405 nm**
Track pitch	1.60 µm	0.74 µm	**0.30 µm**
Capacity	about 700 MB	4.7 GB (single layer)	25 GB (single), 50 GB (dual)
Typical use	Music, small files	Standard-definition films, games	HD films, games

Dual layering: two recording layers on one side, nearly **doubling the capacity**; the laser focuses on each layer separately.

Optical discs are used for **backup, distributing software and films**, and transferring files.

11. Virtual Memory

Virtual memory = RAM + **swap space** on the HDD/SSD. It is used when **RAM is full**.

1. The OS finds the **oldest or least-used data (pages)** in RAM.
2. These pages are **moved to the swap space** on the HDD/SSD.
3. The freed RAM is used by the **new program**.
4. When the moved data is needed again, it is **swapped back** into RAM.

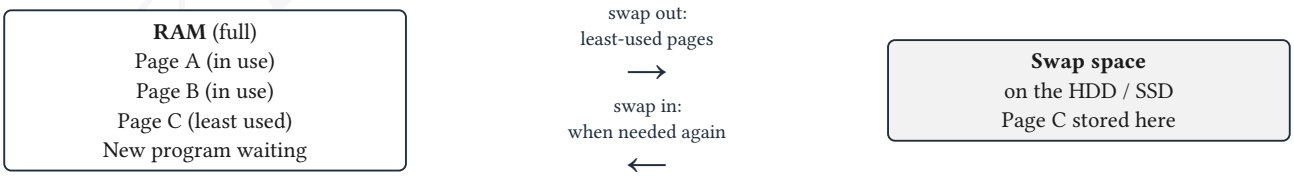


Figure 18: Virtual memory: pages are swapped between RAM and the swap space

Paging: data is moved in **fixed-size blocks called pages**.

Benefits: programs **larger than RAM** can still run; avoids crashes when RAM is full; reduces the need to buy more RAM.

Drawback: slower, because secondary storage is much slower than RAM.

Disk thrashing: so much swapping between RAM and the HDD that **more time is spent moving data than processing**. The **thrash point** is when the system stops making progress. It can also wear out the HDD.

Reducing thrashing: install more RAM, run **fewer programs** at once, or use an **SSD** instead of an HDD.

12. Cloud Storage

Cloud storage stores data on **remote (off-site) servers** owned by a hosting company, accessed over the **internet**. The same data is kept on **more than one server** (**data redundancy**) so it is still available during repair or failure.

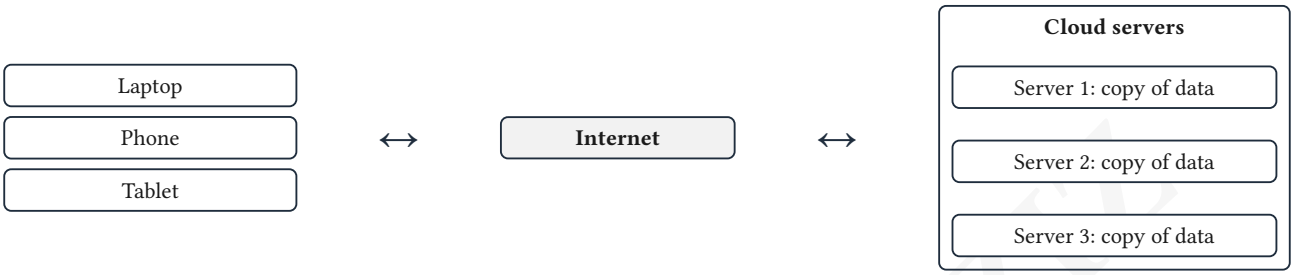


Figure 19: Cloud storage: devices reach copies of the data on remote servers over the internet

Type	Meaning
Public cloud	The **client and the provider are different companies**; storage is shared.
Private cloud	A **dedicated** environment **behind the company's own firewall**; the client and provider are the same organisation.
Hybrid cloud	A mix: **sensitive data in a private cloud**, less sensitive data in a public cloud.

Benefits	Drawbacks
Access files **from any device, anywhere** with internet	Needs a **good internet connection**
No need to carry storage devices	**Costs can be high** for large amounts of storage
Remote backup: helps recover from disk failure or disaster	**Security risk**: hackers, and the user depends on the provider's security
Almost unlimited capacity (can increase easily)	If the **provider fails**, data may be lost

13. Network Interface Card (NIC)

A **NIC** is the hardware that allows a device to **connect to a network**. It is usually built into the device and contains the **MAC address**, which is set at the manufacturing stage. A **wireless NIC (WNIC)** connects using **radio/microwave signals through an antenna**.

14. MAC Address

A **MAC (Media Access Control) address** is a **unique hardware address** of a device's NIC.

- **48 bits**, written as **six pairs of hexadecimal digits**, e.g. `00-1C-B3-4F-25-FF`.
- **First half: manufacturer code**. **Second half: device serial number**.
- Set by the **manufacturer**. If the NIC is replaced, the MAC address changes.

00-1C-B3 **Manufacturer code** (first half)	4F-25-FF **Device serial number** (second half)

Listing 1: Structure of a MAC address

Two types:

- **UAA (Universally Administered Address):** set by the manufacturer; the most common.
- **LAA (Locally Administered Address): changed by the user or organisation** (for example to meet a required format or pass a filter). It must still be **unique** on the network.

15. IP Address

An **IP (Internet Protocol) address** identifies a device's **location on a network or the internet** and is used to **route packets**.

Version	Size and format	Example
IPv4	**32 bits**: four groups of numbers (0–255) separated by dots	`254.25.28.77`
IPv6	**128 bits**: eight groups of hex digits separated by colons	`A8FB:7A88:FFF0:0FFF:3D21:2085:66FB:F0FA`

IPv6 was introduced because IPv4 does **not provide enough addresses**. IPv6 gives many more addresses and **more efficient routing**.

Private and public IP: a **router** assigns each device a **private IP address** (unique only within that network). The router itself has one **public IP address** from the **ISP**, which is unique on the internet.

15.1. Static and Dynamic IP Addresses

Static IP	Dynamic IP
Never changes	**Changes** each time the device connects
Used for **web servers, online databases, FTP servers**	Used by **most home users**
Device is **fully traceable** and the address is **reliable**	**More private** and cheaper; assigned automatically
More expensive to maintain; less privacy	Less reliable for services such as VoIP

DHCP (Dynamic Host Configuration Protocol): a server that **automatically assigns a dynamic IP address** to a device when it connects.

15.2. MAC Address vs IP Address

MAC address	IP address
Physical (hardware) address of the NIC	**Logical** address on a network or the internet
Assigned by the **manufacturer**	Assigned by the **router or ISP** (DHCP)
48 bits in hexadecimal	**32 bits** (IPv4) or **128 bits** (IPv6)
Fixed (UAA) or locally changed (LAA)	**Static or dynamic**
Identifies the **device** within a local network	Used for **routing** between networks

16. Router

A **router connects different networks** (e.g. a LAN to the internet/WAN) and **forwards data packets** between them.

- It reads the **destination IP address** in each packet and sends it on the **best route**.
- It **assigns private IP addresses** to devices on the LAN, and has a **public IP address** for the internet.
- It can be **wired or wireless**, and usually sits behind a **firewall**.
- Within the LAN, the packet is delivered to the correct device using its **MAC address**.

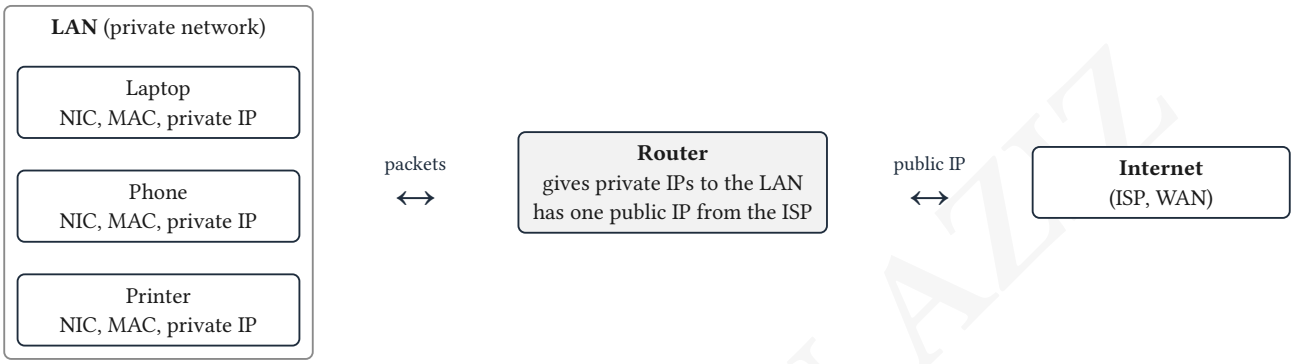


Figure 20: A router connecting a home LAN to the internet

> **Exam Tip:** Router = connects **different networks** using **IP addresses**. NIC and MAC address = identify **one device**.

17. Model Answers

Describe the Fetch–Decode–Execute cycle.
- The address in the PC is copied to the MAR.
- The instruction at that address is copied to the MDR, then to the CIR.
- The PC is incremented by 1.
- The control unit decodes the instruction and executes it by sending control signals.

Explain how cache improves CPU performance.
- Cache stores frequently used data and instructions inside the CPU.
- It is faster to access than RAM, so the CPU waits less.

Give two risks of overclocking.
- The CPU may overheat.
- Instructions may not finish in time, so the computer becomes unstable and crashes.

Explain how a barcode is read.
- A red laser or LED is passed over the barcode, and the dark bars reflect less light.
- Sensors detect the reflected light and the pattern is converted to digital data.
- The number is looked up in the database to find the price.

Give two advantages of SSD over HDD.
- No moving parts, so it is more reliable.
- Faster data access and lower power consumption.

Explain virtual memory.
- When RAM is full, the least-used data is moved from RAM to swap space on the HDD/SSD.

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- This frees RAM for the new program, and data is swapped back when needed.

Describe how a sensor and microprocessor control a greenhouse heater.
- The temperature sensor sends readings to the ADC, which converts them to digital.
- The microprocessor compares them with the stored value.
- If the temperature is too low, it sends a signal to switch the heater on; if too high, to switch it off.
- The sensor keeps reading and the process repeats (feedback).

Give three differences between a MAC address and an IP address.
- A MAC address is set by the manufacturer; an IP address is assigned by the router or ISP.
- A MAC address is 48 bits; an IP address is 32 bits (IPv4) or 128 bits (IPv6).
- A MAC address identifies the device hardware; an IP address is used to route data between networks.

Describe how a QR code can be read.
- The phone camera takes an image of the code.
- The app converts the squares into data.
- The browser opens the web address contained in the code.

18. Common Exam Questions

- Name the registers (PC, MAR, MDR, CIR, ACC) and state their functions.
- Describe the Fetch–Decode–Execute cycle, or state the role of the buses.
- Explain how clock speed, cores, cache and bus width affect performance.
- Explain opcode, operand and instruction set.
- Define an embedded system and give examples, benefits and drawbacks.
- Describe how a barcode scanner, QR code, touch screen, 2D/3D scanner or microphone works.
- Compare inkjet and laser printers, or DLP and LCD projectors.
- Explain the use of a 3D printer and give two uses.
- Compare LED, LCD and OLED screens.
- Name sensors and describe how they are used in monitoring or control (feedback, ADC, DAC).
- Compare RAM and ROM, DRAM and SRAM, HDD and SSD, DVD and Blu-ray.
- Explain virtual memory, thrashing, and cloud storage (benefits and drawbacks).
- Explain NIC, MAC address, IP address (static and dynamic) and the router.

19. Quick Revision

- **CPU** = CU + ALU + registers. **Von Neumann:** programs and data stored in memory; instructions run one after another.
- **PC → MAR → MDR → CIR**, then PC + 1. **ACC** stores ALU results.
- **Address bus:** one way. **Data bus** and **control bus:** two ways.
- **Performance:** higher clock speed, more cores, larger cache, wider buses. **Overclocking** causes overheating and crashes.
- **Opcode** = operation, **operand** = data. **Embedded system** = dedicated hardware + software for one job.
- **Barcode:** laser reflection. **QR code:** more data, error correction, read by camera. **CCD + ADC** in cameras and scanners.
- **Capacitive / infrared / resistive** touch screens. **Inkjet:** small jobs, colour. **Laser:** large jobs, fast. **OLED:** no backlight.

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- **Sensor → ADC → microprocessor → (DAC) → actuator. Monitoring** only watches; **control** acts and uses **feedback**.
- **RAM:** volatile, read/write. **ROM:** non-volatile, read only, holds start-up instructions. **SRAM** (cache, fast) vs **DRAM** (main memory, refreshed).
- **HDD:** magnetic, latency. **SSD:** no moving parts, fast, flash. **Optical:** pits and lands, blue laser = highest capacity.
- **Virtual memory:** swap space on disk when RAM is full; too much swapping = **thrashing**.
- **Cloud:** remote servers, access anywhere, depends on internet and security. **Public, private, hybrid.**
- **MAC:** 48-bit hardware address from the manufacturer. **IP:** IPv4 32-bit, IPv6 128-bit; static or dynamic (DHCP). **Router** joins networks.

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